

# Dose–Response Relationship of Prenatal Mercury Exposure and IQ: An Integrative Analysis of Epidemiologic Data

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Background Prenatal exposure to mercury has been associated with adverse childhood neurologic outcomes in epidemiologic studies. Dose–response information for this relationship is useful for estimating benefits of reduced mercury exposure. Objectives We estimated a dose–response relationship between maternal mercury body burden and subsequent childhood decrements in intelligence quotient (IQ), using a Bayesian hierarchical model to integrate data from three epidemiologic studies. Methods Inputs to the model consist of dose–response coefficients from studies conducted in the Faroe Islands, New Zealand, and the Seychelles Islands. IQ coefficients were available from previous work for the latter two studies, and a coefficient for the Faroe Islands study was estimated from three IQ subtests. Other tests of cognition/achievement were included in the hierarchical model to obtain more accurate estimates of study-to-study and end point-to-end point variability. Results We find a central estimate of  $-0.18$  IQ points (95% confidence interval,  $-0.378$  to  $-0.009$ ) for each parts per million increase of maternal hair mercury, similar to the estimates for both the Faroe Islands and Seychelles studies, and lower in magnitude than the estimate for the New Zealand study. Sensitivity analyses produce similar results, with the IQ coefficient central estimate ranging from  $-0.13$  to  $-0.25$ . Conclusions IQ is a useful end point for estimating neurodevelopmental effects, but may not fully represent cognitive deficits associated with mercury exposure, and does not represent deficits related to attention and motor skills. Nevertheless, the integrated IQ coefficient provides a more robust description of the dose–response relationship for prenatal mercury exposure and cognitive functioning than results of any single study.